INCH-POUND

MIL-T-27/295C
8 June 2012
SUPERSEDING
MIL-T-27/295B
20 April 1993

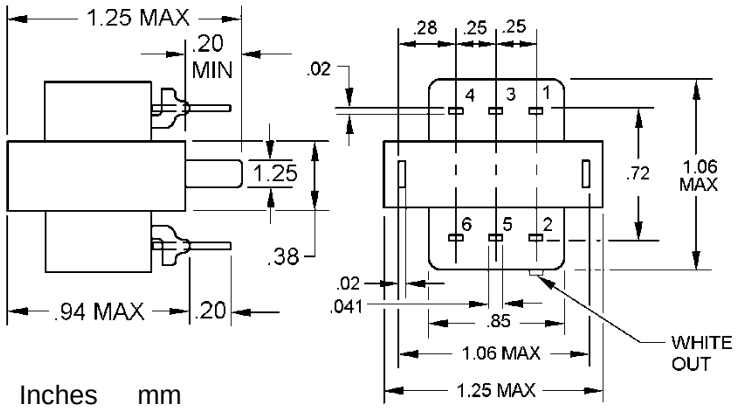
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-DOWN

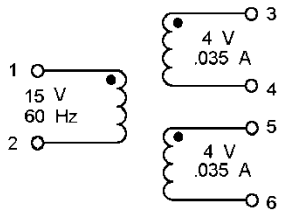
Inactive for new design after 20 April 1993.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the products described herein shall consist of this document and the latest issue of specification MIL-PRF-27.



Inches	mm
.02	0.5
.041	1.04
.125	3.18
.20	5.1
.25	6.4
.28	7.1
.38	9.7
.72	18.3
.85	21.6
.94	23.9
1.06	26.9
1.25	31.8



WV: 443 V MAX
ALTITUDE: 10,000 FT MAX

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three-place decimals and ±.03 (0.8 mm) for two-place decimals.
4. Tolerance on terminal position dimensions is ±.015 (0.38 mm).
5. Marking shall be on the top of the case.
6. Terminal numbers are not required. Terminal number 2 is indicated by white dot as shown.

FIGURE 1. Dimensions and configurations.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the windings.)

Electrical ratings:

Primary voltage (1-2): 115 volts rms, 60 Hz.

Primary current (1-2): 0.035 ampere.

Volt ampere: .66 voltampere.

Secondary:

(3-4): 4 volts, .035 ampere rms.
(5-6): 4 volts, .035 ampere rms.

Working voltage: 443 volts, maximum.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: Solder type.

Material: Brass, tinned or silver plated.

Height: .20 ±.03 inch.

Width: .041 ±.005 inch.

Thickness: .02 ±.03 inch.

Weight: 56 grams, maximum.

Operating temperature range: +55°C to +105°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage (at sea level): 1,240 volts rms.

Electrical characteristics:

No load: With 15 volts rms, 60 Hz across (1-2), current in (1-2) shall not exceed .024 ampere and the power in (1-2) shall not exceed .100 watts.

Voltage across (3-4) and (5-6): 7.65 volts ±5 percent.

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Rated load: With 15 volts rms, 60 Hz across (1-2) and 35 milliamperes resistive load in (3-4) and (5-6), the voltage across (3-4) and (5-6) shall be 4 volts ±10 percent.

DC resistance:

(1-2): 157 ohms ±20 percent.
(3-4): 28.1 ohms ±20 percent.
(5-6): 27.9 ohms ±20 percent.

Polarity: Additive, with terminals 2, 3 and 5 connected.

Temperature rise: 40°C maximum with rated voltage and current at an ambient temperature of 65°C.

Shock (high impact): MIL-STD-202, method 207.

VERIFICATION:

Qualification inspection: Not applicable for this specification sheet.

Conformance inspection: Groups A and B tests of MIL-PRF-27 shall be applicable.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/295-01.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA – CC

(Project 5950-2009-084)

Review activities:
Army – AR, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.daps.dla.mil.

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